```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

print("Jupyter Notebook is working!")
```

```
Jupyter Notebook is working!
```

```python
df = pd.read_csv("small_ex_data.csv")

df.head(10)
```

```
          Order ID  Order Date      Customer Name  Country  
State  \
0  KL-2018-5607031  01-01-2018        Lani Bowers  Austria  
Vienna   
1  KL-2018-5607032  01-01-2018        Lana Adkins  Austria  
Vienna   
2  KL-2018-5607033  01-01-2018      Sopoline Noel  Austria  
Vienna   
3  KL-2018-5607034  02-01-2018       Abbot Horton  Germany  Lower 
Saxony   
4  KL-2018-5607035  02-01-2018   Hedda Livingston  Germany  Lower 
Saxony   
5  KL-2018-5607036  02-01-2018       Henry Hodges  Germany  Lower 
Saxony   
6  KL-2018-5607037  02-01-2018  Aristotle Barrett  Germany  Lower 
Saxony   
7  KL-2018-5607038  02-01-2018         Wendy Neal  Germany  Lower 
Saxony   
8  KL-2018-5607039  03-01-2018   Raymond Mckenzie  Denmark  
Hovedstaden   
9  KL-2018-5607040  03-01-2018      Zephr Hickman    Spain  
Valenciana   

         City   Region  Sales  Profit  
0      Vienna  Central   2934    1819  
1      Vienna  Central   4354    2787  
2      Vienna  Central   2102    1534  
3      Langen  Central   3261    2446  
4      Langen  Central   2577    1907  
5      Langen  Central   3131    1973  
6      Langen  Central   2683    1798  
7      Langen  Central   2613    1751  
8  Copenhagen    North   4076    2609  
9      Gandia    South   4400    3432  
```

Basice Information

```
df.shape
```

```
(18, 9)
```

```
df.columns
```

```
Index(['Order ID', 'Order Date', 'Customer Name', 'Country', 'State',
'City',
       'Region', 'Sales', 'Profit'],
      dtype='str')
```

```
df.isnull().sum()
```

```
Order ID         0
Order Date       0
Customer Name    0
Country          0
State            0
City             0
Region           0
Sales            0
Profit           0
dtype: int64
```

```
df.dtypes
```

```
Order ID           str
Order Date         str
Customer Name      str
Country            str
State              str
City               str
Region             str
Sales            int64
Profit           int64
dtype: object
```

```
df.describe()
```

```
             Sales       Profit
count    18.000000    18.000000
mean   3154.222222  2232.944444
std     714.166775   577.199168
min    2102.000000  1325.000000
25%    2630.500000  1803.250000
50%    3060.000000  2211.000000
75%    3413.000000  2593.000000
max    4400.000000  3432.000000
```

Numpy Operation

```
sales=np.array(df["Sales"])
profit=np.array(df["Profit"])

np.sum(sales)

np.int64(56776)

np.min(sales)

np.int64(2102)

np.max(sales)

np.int64(4400)

np.mean(sales)

np.float64(3154.222222222222)

np.sum(profit)

np.int64(40193)

np.mean(profit)

np.float64(2232.9444444444443)
```

PANDAS OPERATIONS

```
df.nlargest(5,"Sales")

           Order ID  Order Date     Customer Name  Country
State  \
9   KL-2018-5607040  03-01-2018     Zephr Hickman    Spain
Valenciana
1   KL-2018-5607032  01-01-2018       Lana Adkins  Austria
Vienna
12  KL-2018-5607043  04-01-2018     Dahlia Porter    India
Bihar
8   KL-2018-5607039  03-01-2018  Raymond Mckenzie  Denmark
Hovedstaden
17  KL-2018-5607048  05-01-2018   Quinlan Holcomb  Belgium  Flemish
Brabant

          City   Region  Sales  Profit
9       Gandia    South   4400    3432
1       Vienna  Central   4354    2787
12       Patna    South   4216    3162
```

```
8   Copenhagen    North   4076    2609
17      Leuven  Central   3463    2770
```

```
df.nlargest(5,"Profit")
```

```
              Order ID  Order Date     Customer Name  Country
State  \
9   KL-2018-5607040  03-01-2018     Zephr Hickman    Spain
Valenciana
12  KL-2018-5607043  04-01-2018     Dahlia Porter    India
Bihar
1   KL-2018-5607032  01-01-2018       Lana Adkins  Austria
Vienna
17  KL-2018-5607048  05-01-2018   Quinlan Holcomb  Belgium  Flemish
Brabant
8   KL-2018-5607039  03-01-2018  Raymond Mckenzie  Denmark
Hovedstaden

          City   Region  Sales  Profit
9       Gandia    South   4400    3432
12       Patna    South   4216    3162
1       Vienna  Central   4354    2787
17      Leuven  Central   3463    2770
8   Copenhagen    North   4076    2609
```

```
df.groupby("Region")["Sales"].sum()
```

```
Region
Central    36316
North       6542
South      13918
Name: Sales, dtype: int64
```

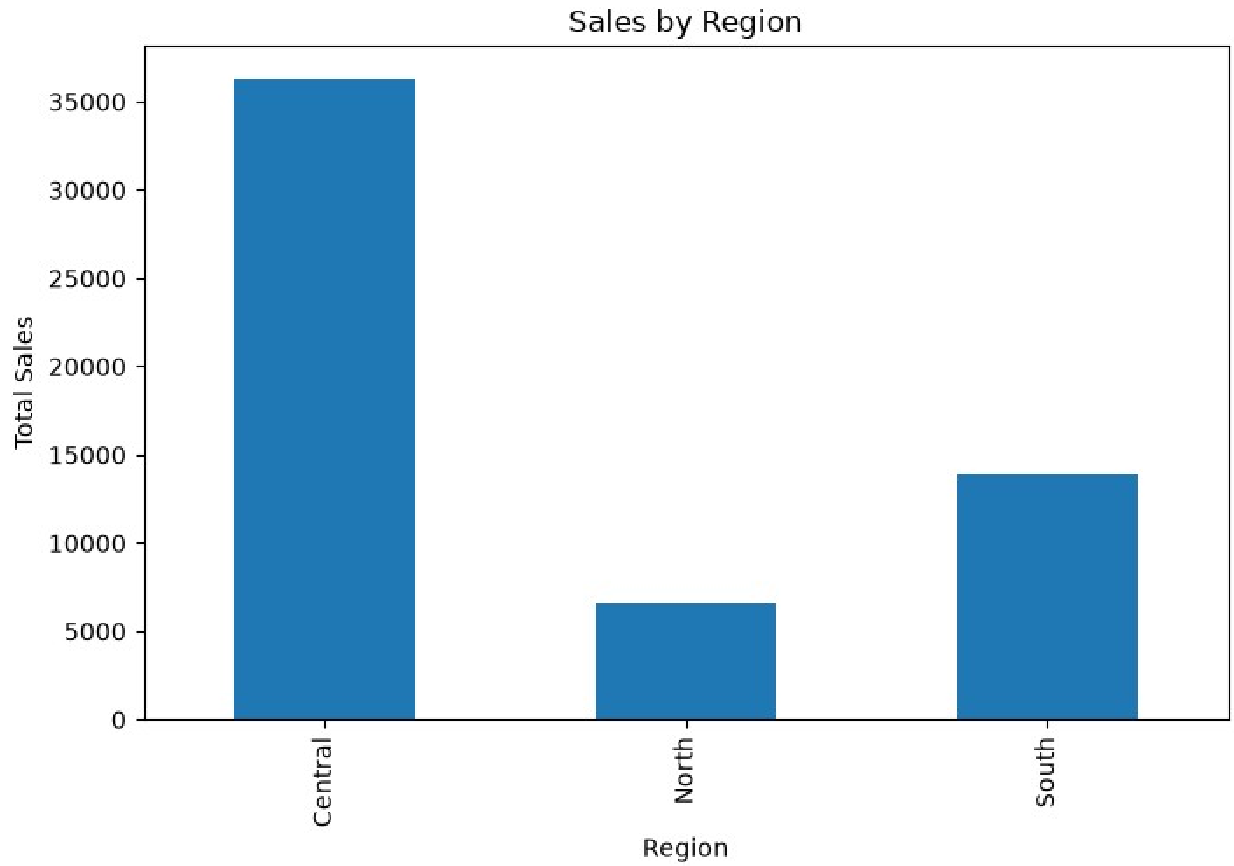
```
df.groupby("Region")["Profit"].sum()
```

```
Region
Central    25752
North       4212
South      10229
Name: Profit, dtype: int64
```

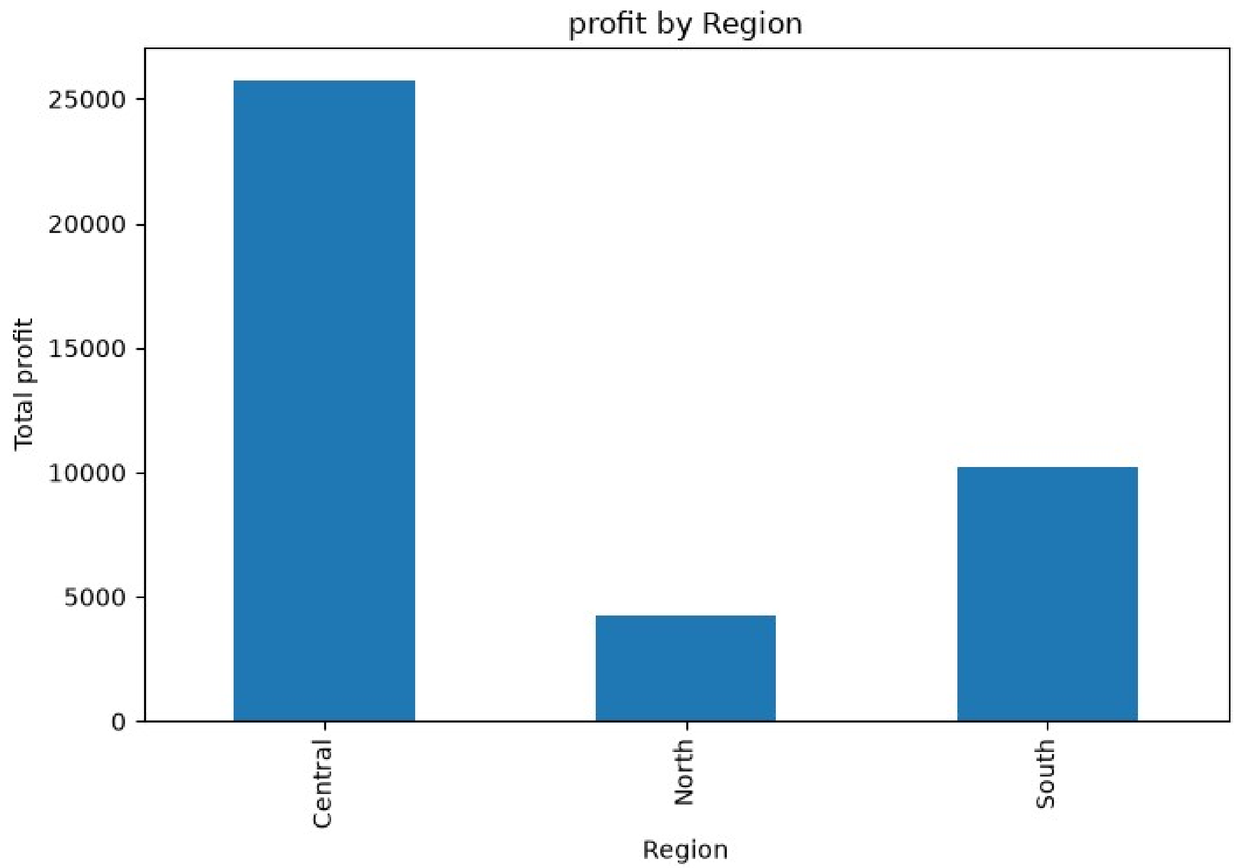
MATPLOTLIB VISUALIZATIONS

```
rigion_sales = df.groupby("Region")["Sales"].sum()

plt.figure(figsize=(8,5))
rigion_sales.plot(kind="bar")
plt.title("Sales by Region")
plt.xlabel("Region")
```

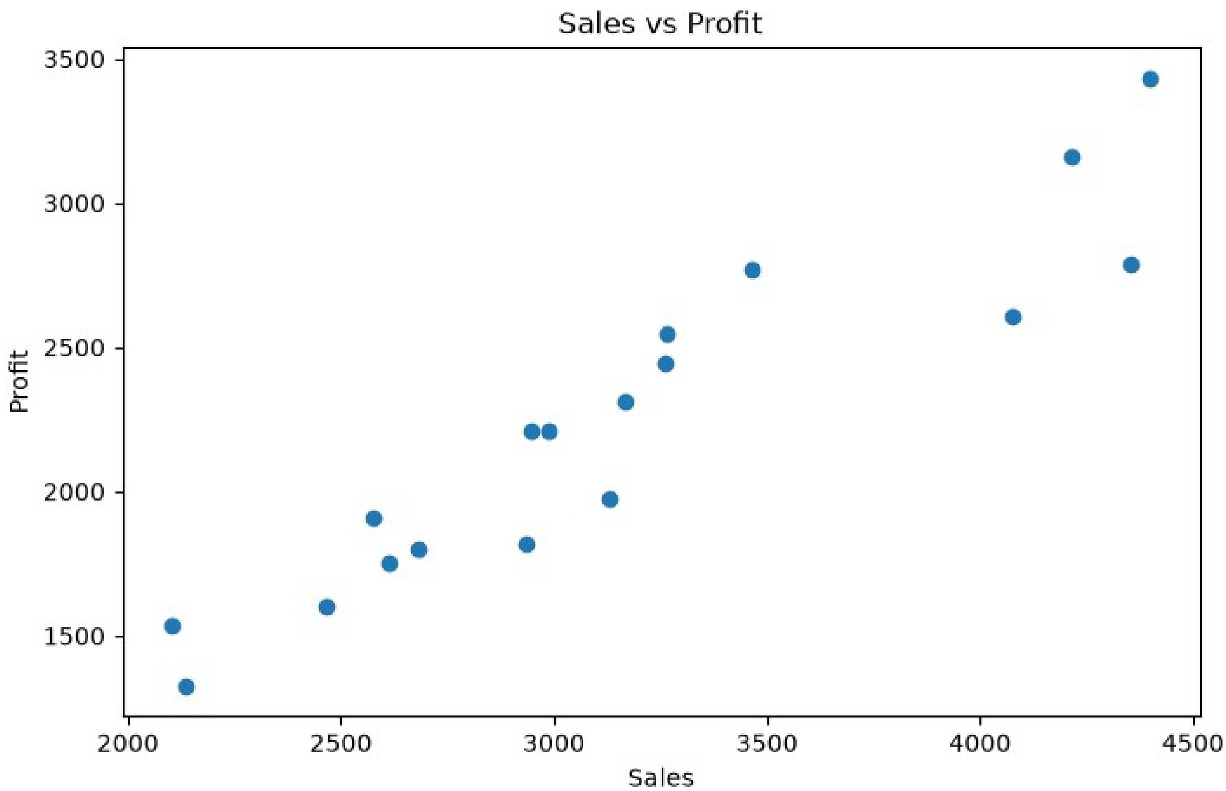
```
plt.ylabel("Total Sales")
plt.show()
```



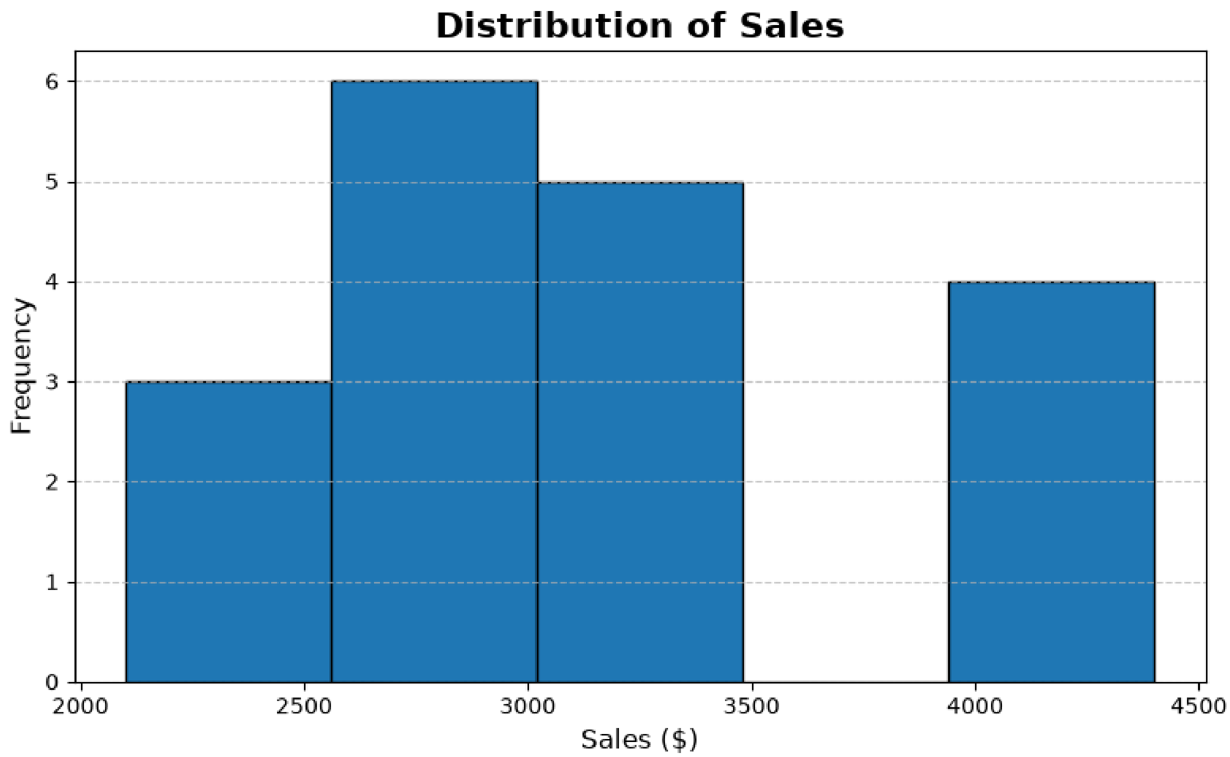

```
region_profit=df.groupby("Region")["Profit"].sum()

plt.figure(figsize=(8,5))
region_profit.plot(kind="bar")
plt.title("profit by Region")
plt.xlabel("Region")
plt.ylabel("Total profit")
plt.show()
```



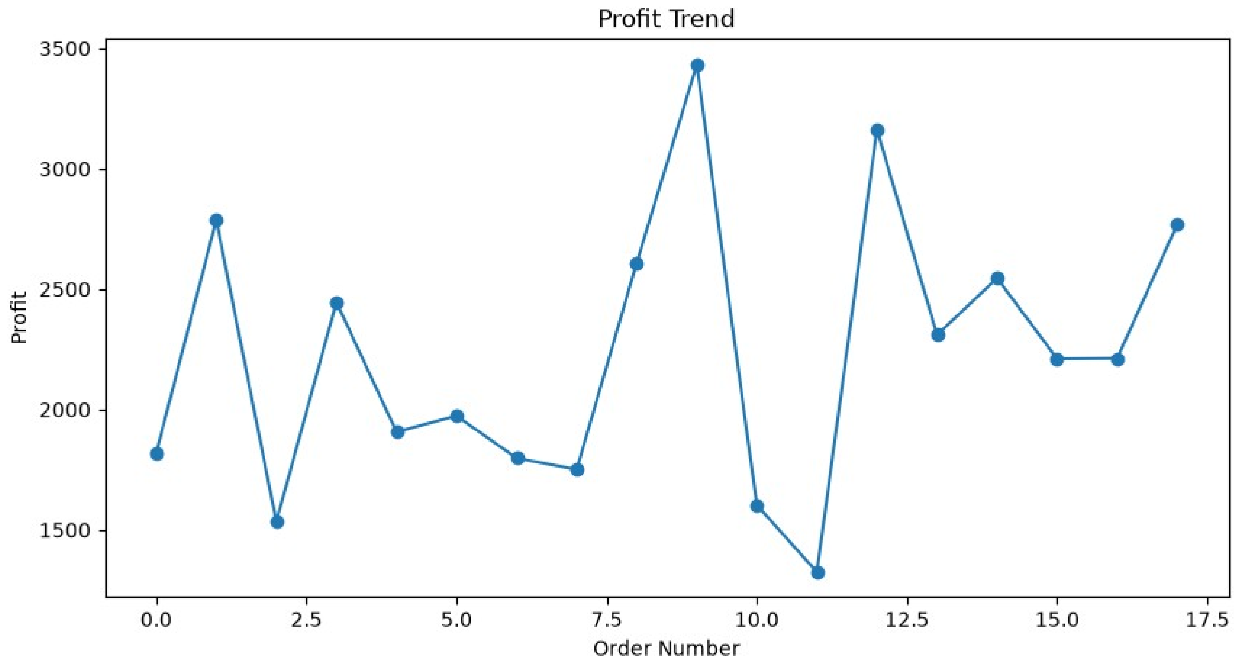

```
plt.figure(figsize=(8,5))
plt.scatter(df["Sales"], df["Profit"])
plt.title("Sales vs Profit")
plt.xlabel("Sales")
plt.ylabel("Profit")
plt.show()
```



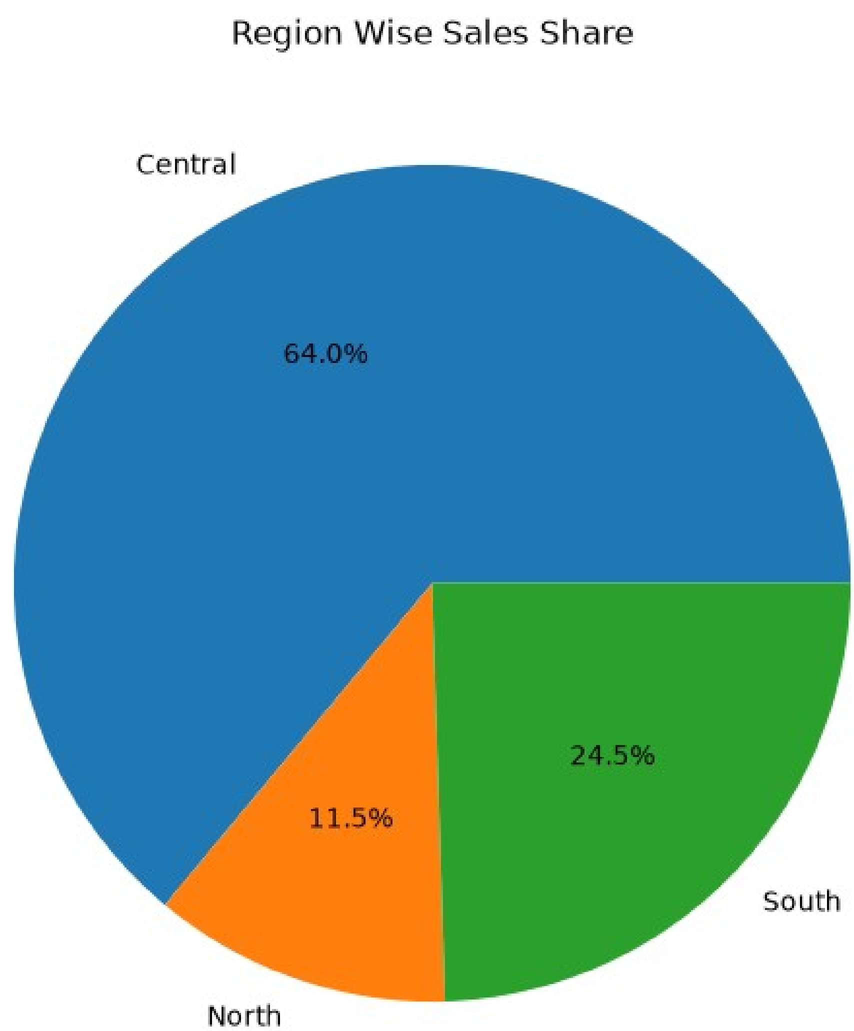

```
plt.figure(figsize=(8,5))

plt.hist(df["Sales"], bins=5, edgecolor='black')

plt.title("Distribution of Sales", fontsize=16, fontweight='bold')
plt.xlabel("Sales ($)", fontsize=12)
plt.ylabel("Frequency", fontsize=12)

plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.tight_layout()
plt.show()
```




```
plt.figure(figsize=(10,5))
plt.plot(df["Profit"], marker='o')
plt.title("Profit Trend")
plt.xlabel("Order Number")
plt.ylabel("Profit")
plt.show()
```




```
plt.figure(figsize=(7,7))
rigion_sales.plot(kind="pie", autopct='%1.1f%%')
plt.title("Region Wise Sales Share")
plt.ylabel("")
plt.show()
```


Region Wise Sales Share
Central
64.0%
11.5%
24.5%
North
South